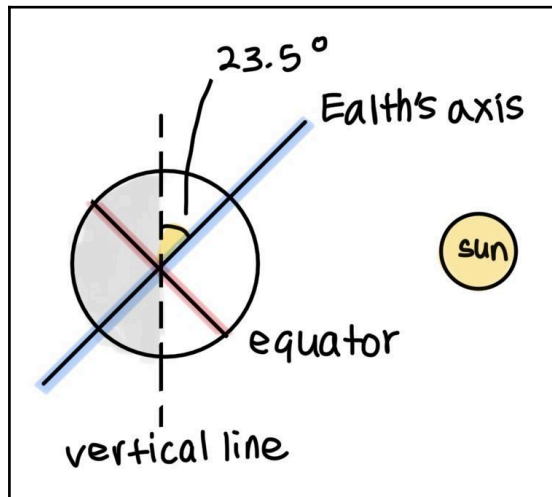


6.1 The Earth and the Solar system

The Earth:

- It is a rocky planet that orbits the sun, in a nearly circular orbit



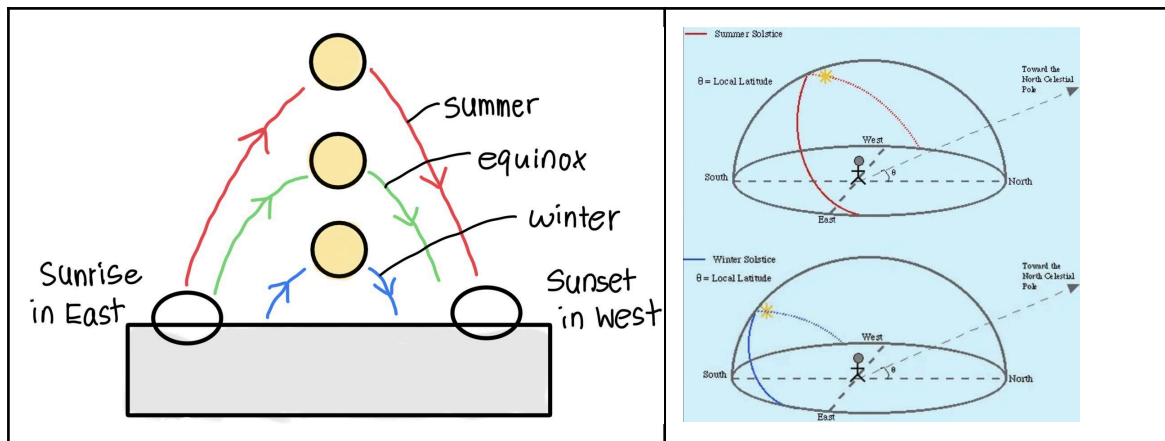
- It rotates from West to East on its axis
- The axis is tilted at an angle of 23.5°
- It completes 1 full rotation in 24 hours (1 day)

The earth's rotation on its axis creates:

- The sun rising in the East, and setting in the West
- So, the periodic cycle of day and night

Rising and setting of the Sun

- Note -> For southern hemisphere the seasons are switched, so summer in northern hemisphere is winter in southern hemisphere



During Summer:

- The Sun rises North of East and sets North of West
- Midday reaches its highest point in the sky
- Daylight hours reach their maximum (during summer's solstice)

During Winter:

- The Sun rises South of East and sets South of West
- Midday reaches its lowest point in the sky
- Daylight hours reach their minimum (during winter's solstice)

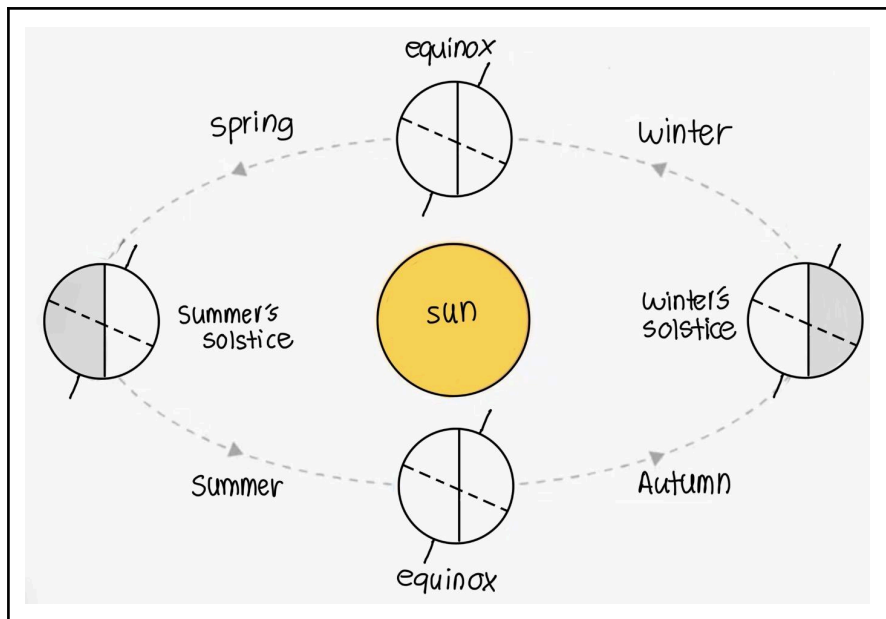
During Equinox:

- The sun rises exactly in the East and sets exactly in the West
- Day and night are roughly equal in length

The Earth's orbit and its seasons

- The Earth orbits the Sun once in 365 days (1 year)
- The orbiting of the Earth around the sun, and the Earth's axis tilt and rotation creates the seasons

For the Northern Hemisphere:



During Spring and Summer:

- The Northern hemisphere is tilted towards the sun
- There is a longer day than night

At Summer's solstice: (Start of Summer)

- Northern hemisphere has its longest day

During Autumn and Winter:

- The Northern hemisphere is tilted away from the sun
- There is a shorter day than night

At Winter's solstice: (Start of Winter)

- Northern hemisphere has its shortest day

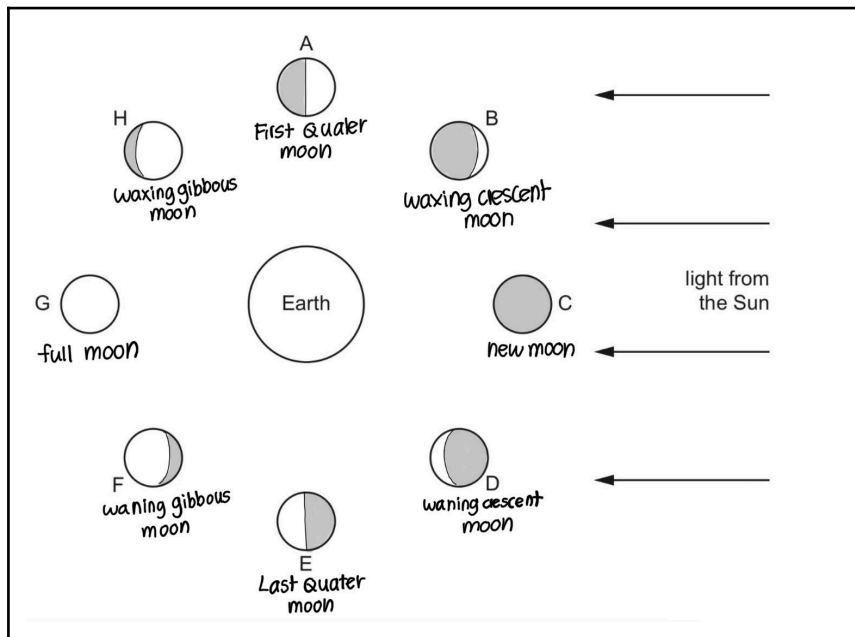
At Equinoxes: (Start of Autumn and Spring)

- Night and day are equal in both hemispheres

Moon and Earth

- The moon is a natural satellite of Earth
- It orbits the earth once every 28 days
- The moon also rotates on its axis once every 28 days, which is why the same side is always faced toward Earth

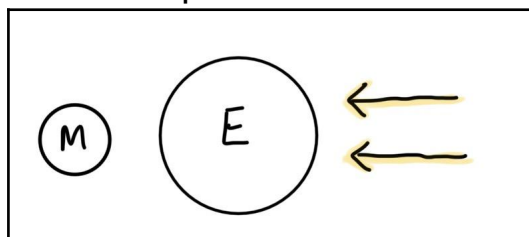
Phases of the moon:



Note:

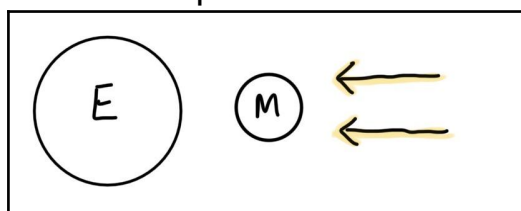
- The circles show how the moon appears from Earth at different positions in its orbit
- Half of the moon is always illuminated by the sun

Full moon phase:



- The Earth is between the moon and the sun
- The side of the moon facing the earth is completely illuminated by the sun, making it appear completely lit

New moon phase:



- The Moon is between the earth and the sun
- The side of the moon facing Earth is not illuminated at all, which makes it appear invisible

Crescent shapes:

- Moon is partially illuminated by sunlight
- The side of the moon facing Earth is mostly not illuminated, which makes it appear slightly lit, crescent shaped

Gibbous shapes:

- Moon is partially illuminated by sunlight
- The side of the moon facing Earth is mostly illuminated, which makes it appear mostly lit

Orbital speed

- When planets or moons orbit another planet their paths, are often nearly circular

Average orbital speed:

For 1 orbit:

- Distance = $2\pi r$ (circumference, using orbital radius)
- Time = time taken for one orbit (orbital period)

$$v = \frac{2\pi r}{t}$$

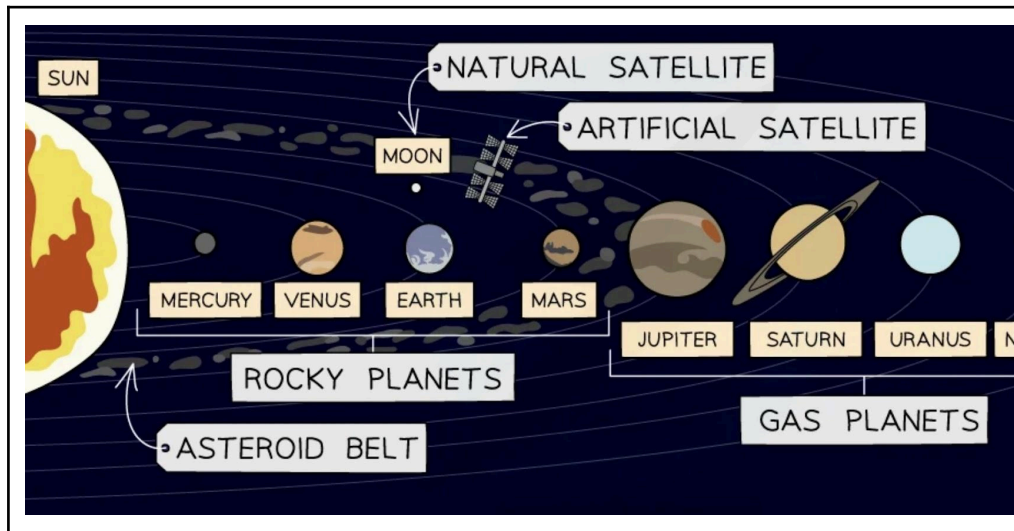
(average orbital speed (m/s)) = $\frac{2\pi r$ (circumference (m))}{t (time (s))}

(orbital radius (m))

- r represents average radius of the orbit
- t represents orbital period

The Solar System

- There are many planetary systems in the universe, with planets orbiting a host star
- Our planetary system is called the Solar System, and it is located in the Milky Way galaxy



The Solar system contains:

- One star, the Sun
- 8 planets orbiting the sun
- Minor planets that orbit the sun -> dwarf planets such as Pluto, and asteroids in the asteroid belt
- Smaller solar system bodies -> comets and satellites (natural + artificial)
- Moons that orbit the planets (natural satellites)

- The first 4 planets - Mercury, Venus, Earth and Mars are small, rocky planets
 - The last 4 planets - Jupiter, Saturn, Uranus and Neptune are large, gaseous planets
 - The asteroid belt lies between the Mars and Jupiter
-

The Sun:

- The Sun lies at the center of the solar system
- It contains most of the mass of the solar system, which is why planets orbit the sun
- The force that keeps an object in orbit around the sun, is the gravitational attraction around the sun

As the distance from the sun increases,

- The Sun's gravitational field strength decreases
- The orbital speeds of the planet decreases

The strength of a gravitational field:

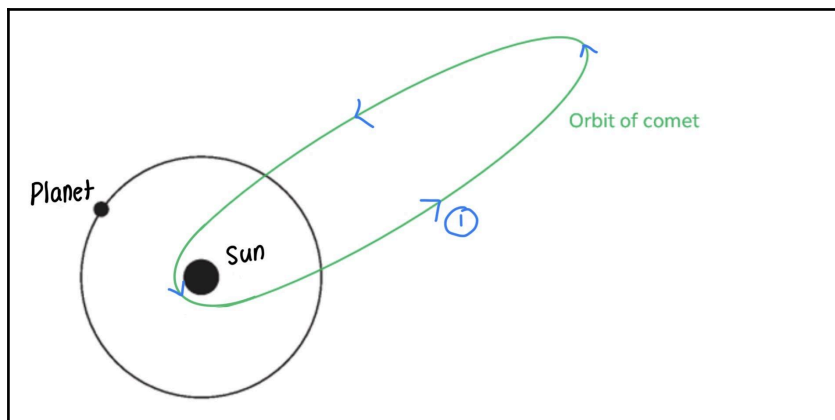
- At the surface of a planet, depends on the mass of the planet ->
 $w = mg$
- Around a planet, decreases as the distance from the planet decreases

Planets, minor planets and comets

- They have elliptical orbits -> the sun is not at the center of the elliptical orbit, except when the orbit is approximately circular

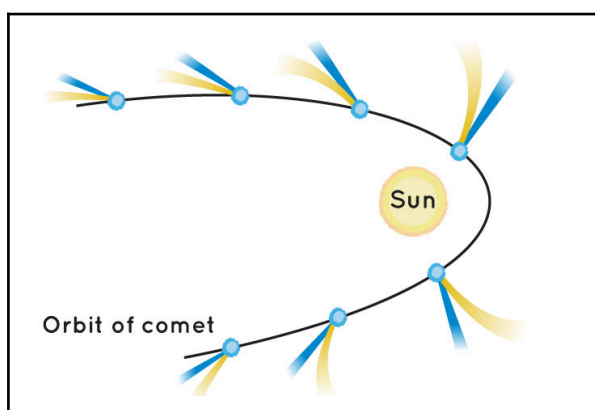
Comets:

- Comets have highly elliptical orbits -> they are more elongated



- As a comet moves further away from the sun, gravitational field strength decreases, causing its speed to decrease
- Kinetic energy is converted to gravitational potential energy
- At the furthest distance away from the sun, gravitational field strength is the weakest, speed is the slowest
- Comet has highest gravitational potential energy and lowest kinetic energy
- The comet's tail is the shortest
- As comet approaches the sun, gravitational field strength increases, causing comet to speed up
- Gravitational potential energy is converted into kinetic energy
- At the closest distance from the sun, gravitational field strength is the strongest, speed is the fastest
- Comet has highest kinetic energy and lowest gravitational potential energy
- The comet's tail is the longest, and most easily seen

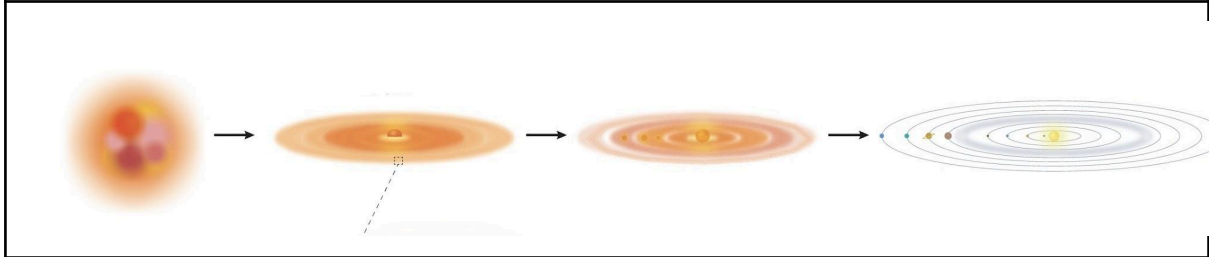
Comet tail:



- As comet approaches the sun, temperature increases.
- Ice melts, forming the comet's tail
- The comet's tail is always directed away from the sun

Accretion model of the solar system

- It is the explanation for how the solar system developed -> It accounts for the differences in the types of planets



- A nebula is an interstellar cloud of gas and dust

Explanation:

- An interstellar cloud of gas and dust collapses due to gravity.
- The materials in the cloud rotate, forming an accretion disk.
- At the same time, this forms a protostar, which later forms the sun (life cycle of star)

- In the accretion disk, tiny dust particles bump into each other and become stuck to each other, forming clumps
- Overtime, these clumps grow bigger, eventually becoming planets

- If the clump was closer to the sun, the temperature was higher
- Only heavy materials such as metals and rocks could remain solid.
- The original nebula contained a small proportion of these heavy elements.
- So, these materials were pulled by gravity to form small and rocky planets

- If the clump was further away from the sun, the temperature was lower
- It was cold enough for lighter materials to condense

- So, these materials were pulled together by gravity to form large and gaseous planets

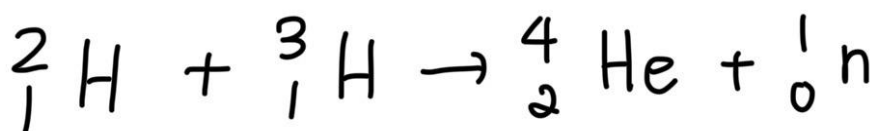
6.2 Stars and the Universe

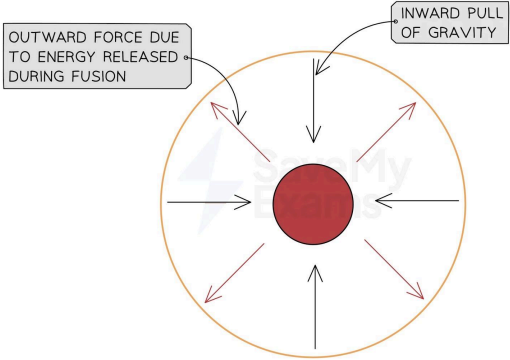
The Sun as a star:

- The Sun is a medium sized star, primarily composed of hydrogen and helium
- It radiates most of its energy in the infrared radiation, visible light, and ultraviolet part of the electromagnetic spectrum

Nuclear fusion in a star:

- Stars are powered by nuclear fusion to release energy
- In the center of a stable star, there is very high pressure and temperature, causing nuclear fusion
- Hydrogen nuclei fuse to form a helium nucleus



 The diagram shows a central red circle representing the core of a star. Surrounding it is a larger orange circle representing the outer layers. Eight arrows point outwards from the core towards the outer edge, labeled 'OUTWARD FORCE DUE TO ENERGY RELEASED DURING FUSION'. Eight arrows point inwards from the outer edge towards the core, labeled 'INWARD PULL OF GRAVITY'. The arrows are of equal length, indicating a balance between the two forces.	- A large amount of energy is released - This energy exerts a force acting outwards on a star, to prevent the star from collapsing under its own gravity.
--	--

The Milky Way:

- Universe is made up of many different galaxies
 - Galaxies are made up of billion of stars
-
- The Sun is a star in the solar system, which is located in the Milky Way galaxy
 - Other stars in the Milky Way galaxy are much further away from Earth, than the Sun is from Earth.
-

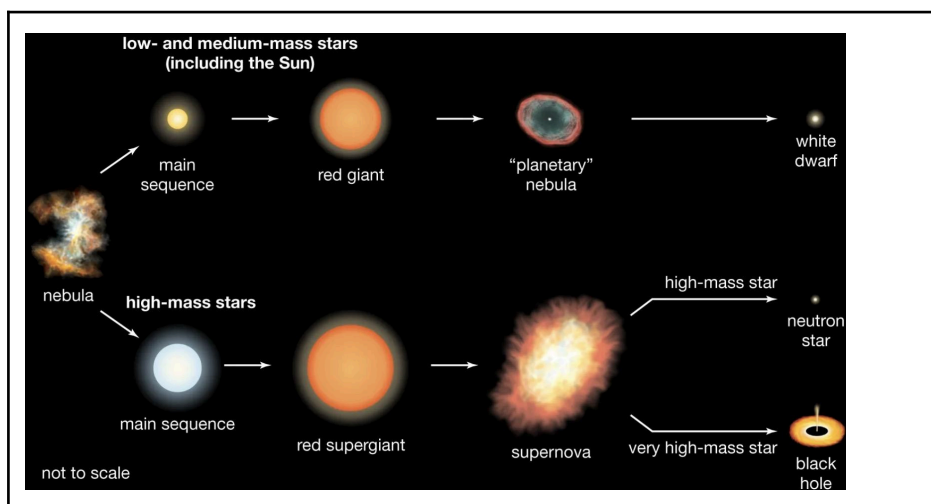
Astronomical distance:

- Can be measured in light years

1 light year: the distance travelled by light in space, in one year

- 1 light year is equal to $9.5 \times 10^{15} \text{ m}$
 - The diameter of the Milky Way galaxy is approximately 100,000 light years
-

Life cycle of a star



- A star is formed from interstellar clouds of gas and dust that contain hydrogen.

Distinguishing between low / medium mass and high mass stars:

- Low/medium mass is when most of the hydrogen in the centre of the star has not been converted to helium
- High mass are is when most of the hydrogen in the centre of the star has been converted to helium

Description:

- An interstellar cloud collapses due to its internal gravitational attraction
- This forms a protostar.
- Nuclear fusion takes place in the protostar. Hydrogen nuclei fuse to form helium nuclei, releasing energy.
- This energy exerts a force acting outwards on the star.
- The inward force of gravitational attraction is balanced by this outwards force, forming a stable star.
- Eventually, all stars run out of hydrogen as fuel for nuclear fusion.

If low and medium mass: (most stars)

- They expand to form red giants
- Red giant forms a planetary nebula, with a white dwarf at its center

If high mass: (more massive)

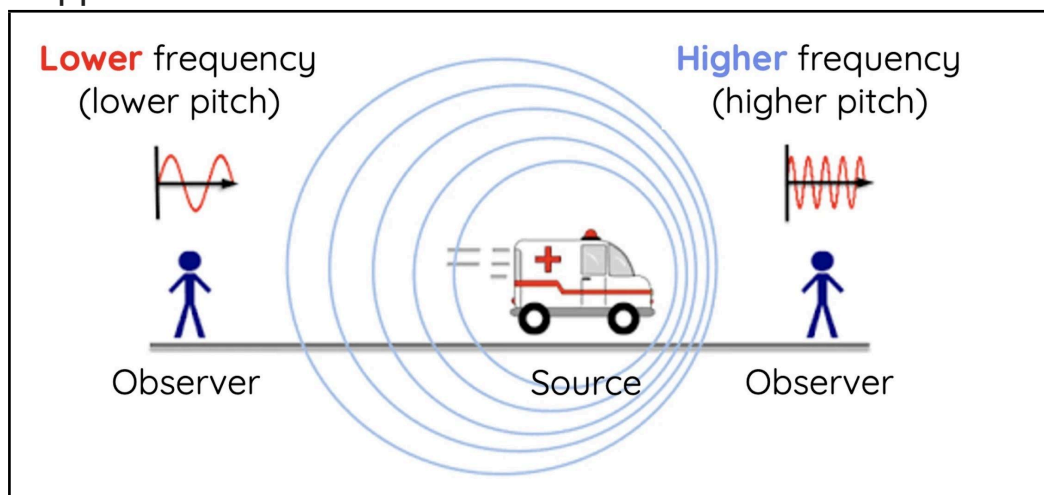
- They expand to form red supergiants
- Red supergiant explodes as a supernova, forming a nebula containing hydrogen and new heavier elements
- It leaves behind a neutron star or black hole at its centre.

- The nebula from a supernova may form new stars with orbiting planets

The Universe

- The universe contains many billions of galaxies

Doppler effect:



- When a stationary ambulance car emits sound waves, the wavefronts spread out symmetrically, forming circles around the source
- When the ambulance car moves, the waves can become compressed in front of it and stretched behind it
 - In front of the source, wavelength decreases, so frequency increases -> pitch increases
 - Behind the source, wavelength increases, so frequency decreases -> pitch decreases

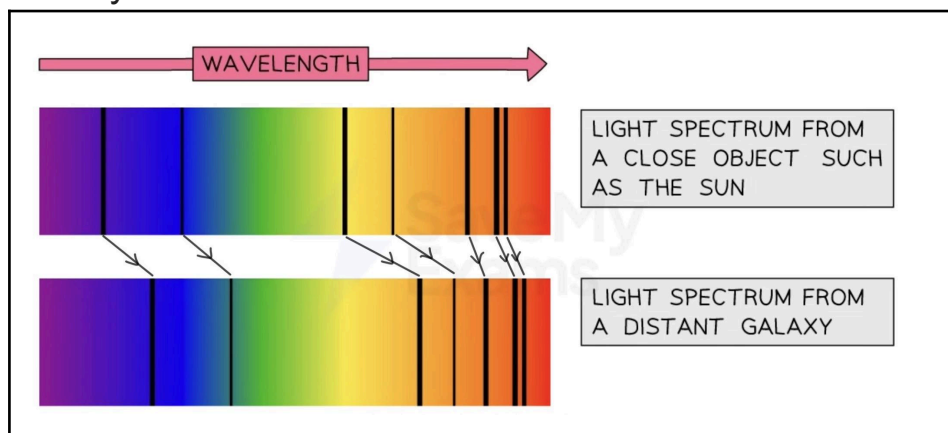
- This is known as Doppler effect - it applies to sound waves and electromagnetic waves (which include light waves)
-

Redshift: an increase in the observed wavelength of the electromagnetic radiation emitted from receding stars and galaxies

- If a source of light moves away from an observer, the observed wavelength increases
 - The wavelength shifts towards the red end of the spectrum
 - This is called 'redshift'

 - If a source of light moves towards an observer, the observed wavelength decreases
 - The wavelength shifts towards the blue end of the spectrum
 - This is called 'blueshift'
-

Galaxy movement:



- This shows that the light from a distant galaxy is redshifted, and shifts to the red end of the spectrum
- This means wavelength is increasing, meaning distant galaxy is moving away from the earth

- Redshift supports the Big Bang Theory -> Redshift in the light from distant galaxies is evidence that the universe is expanding

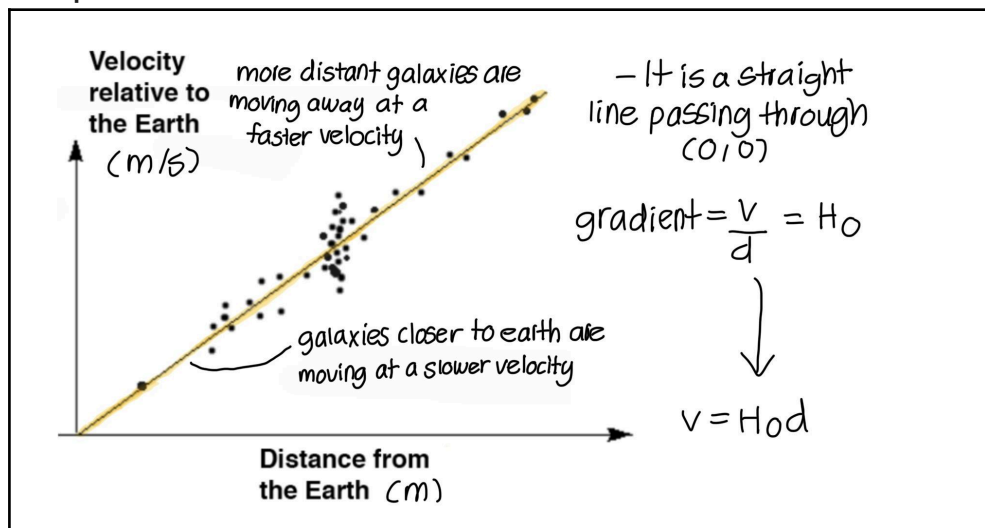
Determining speed and distance:

- The speed of a moving galaxy can be found from the change in wavelength of the galaxy's starlight, due to redshift
- The distance from a galaxy can be determined using the brightness of a supernova in that galaxy

Hubble's law

- Hubble's law: the speed of the galaxies moving away from Earth is directly proportional to their distance from Earth
- This means that the further away the galaxy is, the greater its red shift, and the faster it is moving away from the Earth

Graph:



$$v = H_0 d$$

velocity (m/s)

Hubble constant (s^{-1})

distance (m)

Hubble constant (H_0): the ratio of speed at which galaxy is moving away from Earth, to its distance from Earth

Current estimate for Hubble constant = $2.2 \times 10^{-18} \text{s}^{-1}$

- Hubble's constant tells us how quickly the universe is expanding
-

The Age of the Universe (calculation)

$$t = \frac{d}{v} = \frac{1}{H_0}$$

- This is evidence that all of the matter in the universe was present at a single point
-

Cosmic microwave background radiation (CMBR)

- Microwave radiation of a specific frequency is observed at all points in space around us -> this is known as cosmic microwave background radiation (CMBR)
- CMBR was produced shortly after the universe was formed.
- As the universe expanded, this radiation has been expanded and redshifted into the microwave region of the electromagnetic spectrum.
- This is evidence that the universe has expanded (The Big Bang theory)

CMBR Map

